

ArduPilot Crash Analysis Report

Log: 2026-05-20 15-54-10.bin | Generated: 2026-05-21 05:42

Firmware	Unknown	Hardware	CubeOrangePlus 00170030 3133510E 38393730
Armed At (IST)	20-05-2026 15:54:50 IST	Disarmed At (IST)	—
Flight Duration (Armed → Disarmed)	20-05-2026 15:54:58 IST → 20-05-2026 16:11:19 IST 981s (16.4 min)	Time Reference	IST (India Standard Time, UTC+5:30)
Battery Start	25.08V	Battery Min	21.79V End: 22.20V
Flight Modes	Loiter @15:54:28 IST, Loiter @15:54:48 IST	Log File	2026-05-20 15-54-10.bin

Check Summary

PASS	FAIL	WARN
23	7	4

Crash Verdict

CRASH VERDICT: CONFIRMED CRASH

7 FAIL | 4 WARN | 23 PASS — see Phase 1 & Phase 2 checks below

PHASE 1 – Initial Health Checks

#	Phase	Category	Status	Message	Time Period
1	Phase 1	Clean Boot	WARN	No boot message found — log may be partial or format unrecognised	—
2	Phase 1	Power Stability	WARN	Unloaded voltage drop: 0.21V at 3.2A (t=56.4s, in-flight)	t=56.4s
3	Phase 1	Battery Voltage (Main)	PASS	Voltage nominal: start 25.08V, min 21.79V (6S, warn 21.0V, crit 19.8V)	—
4	Phase 1	IMU Detection	PASS	4 IMUs detected (instances: [0, 1, 2, 100])	—
5	Phase 1	IMU Health	FAIL	IMU100 gyro HW errors: EG max=1016957739 IMU100 accel HW errors: EA max=991432351	—

#	Phase	Category	Status	Message	Time Period
6	Phase 1	IMU FIFO Stability	PASS	No FIFO resets detected	—
7	Phase 1	IMU Vibration	WARN	IMU2 high vibration: 50.0 m/s ² (X:22.9 Y:18.5 Z:50.0)	—
8	Phase 1	IMU Accel Clipping	PASS	No accel clipping events detected	—
9	Phase 1	Compass Detection	PASS	2 compasses detected (instances: [0, 1])	—
10	Phase 1	Magnetic Field	FAIL	MAG0: avg=454mG min=431 max=463 cv=1% [PASS] MAG1: avg=799mG min=431 max=963 cv=5% [FAIL]	—
11	Phase 1	Compass Interference	FAIL	EKF compass variance: SM max=163.49 (0% of flight > threshold 0.5)	—
12	Phase 1	Mag Anomaly	PASS	No mag anomaly events	—
13	Phase 1	GPS Fix	PASS	Fix nominal throughout flight: min=DGPS max=DGPS	—
14	Phase 1	GPS Satellites	PASS	Satellite count good: min=22, avg=27	—
15	Phase 1	GPS Accuracy (HDop)	PASS	Positional accuracy excellent: max HDop=0.64, avg=0.60	—
16	Phase 1	EKF Readiness	PASS	EKF healthy before arm across all 4 lanes	—
17	Phase 1	EKF Variance	PASS	L0: SV=0.15 SP=0.08 SH=0.17 [PASS] L1: SV=0.15 SP=0.08 SH=0.18 [PASS] L2: SV=0.15 SP=0.08 SH=0.19 [PASS]	—
18	Phase 1	EKF Lane Switch	PASS	No in-flight EKF lane switching	—
19	Phase 1	EKF Yaw Alignment	WARN	No yaw alignment completion message — GPS yaw or external heading may be in use, or log is partial	—
20	Phase 1	RC Signal	PASS	RC signal present — Protocol: SBUS Active channels: 14 (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14)	—
21	Phase 1	RC Signal Loss	FAIL	RC signal gap(s) while armed: 1 event(s), longest 222.5s at t=63.8s	t=63.8s
22	Phase 1	Arming	PASS	Arming checks passed	t=55.8s
23	Phase 1	Crash Detection	FAIL	MSG: 'Logging @SYS/crash_dump.bin'	t=63.5s
24	Phase 1	Battery Failsafe	PASS	Battery failsafe not triggered	—
25	Phase 1	RC Failsafe	PASS	RC failsafe not triggered	—

System Messages at Boot:

[33.3s] Marut Auto Pilot V1.6.0 (df9e87a2)
 [33.3s] ChibiOS: 88b84600
 [33.3s] CubeOrangePlus 00170030 3133510E 38393730
 [33.3s] Param space used: 1161/5376
 [33.3s] RC Protocol: SBUS
 [33.3s] RCOut: PWM:1-14
 [33.3s] New mission
 [33.3s] New rally

PHASE 2 – Crash Window Analysis

#	Phase	Category	Status	Message	Time Period
1	Phase 2	Initial Mode	PASS	Armed in Loiter — flight modes used: Loiter, Loiter	—
2	Phase 2	Speed	PASS	Speed normal: max=0.7 m/s	—
3	Phase 2	Height	PASS	No abnormal descent (max=1.2m, min=-7.0m)	—
4	Phase 2	GPS Sats	PASS	Satellites stable: min=22, avg=27	—
5	Phase 2	Motor PWM	FAIL	Motors at minimum before disarm (~670 DShot)	t=696.7s
6	Phase 2	Vibration	PASS	Vibration OK: X=8.9 Y=13.1 Z=11.2 m/s ²	—
7	Phase 2	IMU Clipping	PASS	No accel clipping during flight	—
8	Phase 2	Crash Event	FAIL	MSG: 'Logging @SYS/crash_dump.bin'	t=63.5s
9	Phase 2	Motor Fail Flags	PASS	FailFlags=2 only (battery thrust limiting — normal behavior) 6472 records	—

Flight Charts

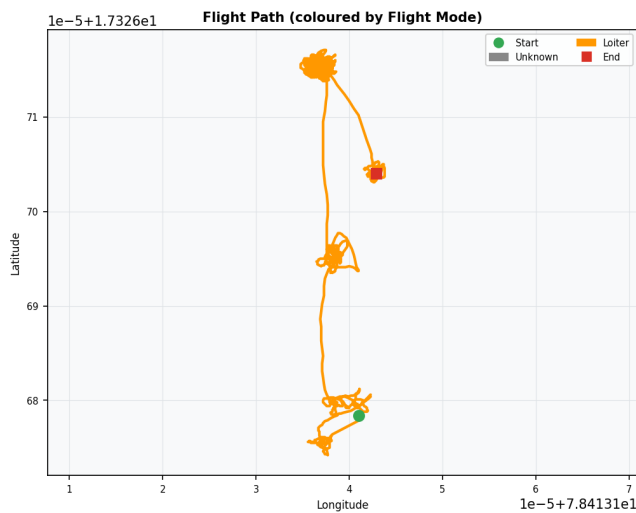
Charts below show full flight data. The red dashed line marks the disarm event; the shaded region marks the 15-second crash window.

Flight Path Maps

GPS route overlaid with sensor data. Each map uses a green→red colour scale where green indicates nominal/safe values and red indicates degraded/critical values.

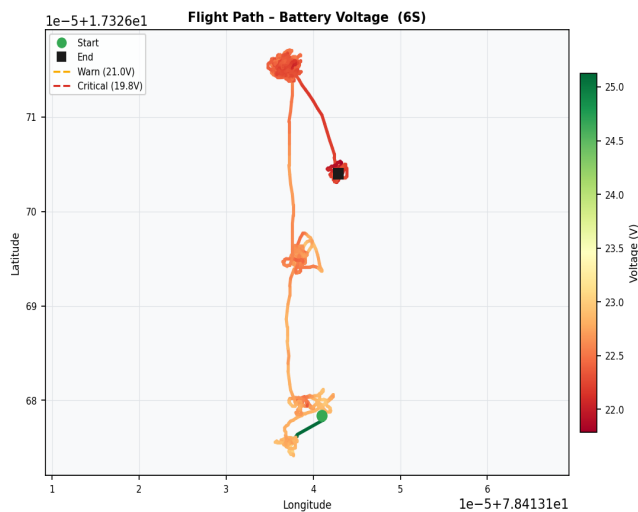
Flight Path – by Mode

GPS route coloured by **flight mode**. ● Start ■ End.



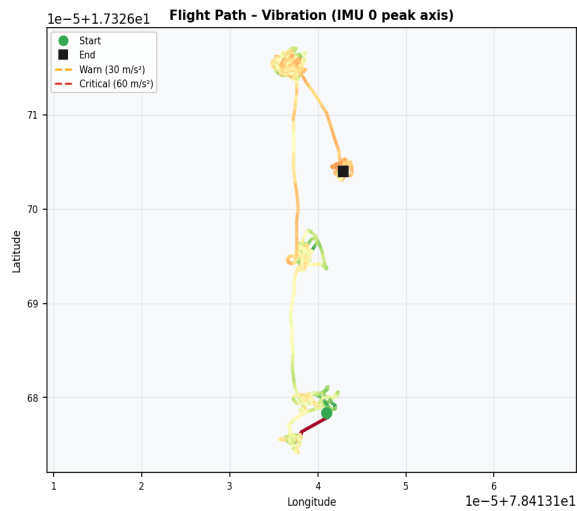
Flight Path – by Voltage

Path coloured by **battery voltage**. Green = high (good) → Red = low (critical).



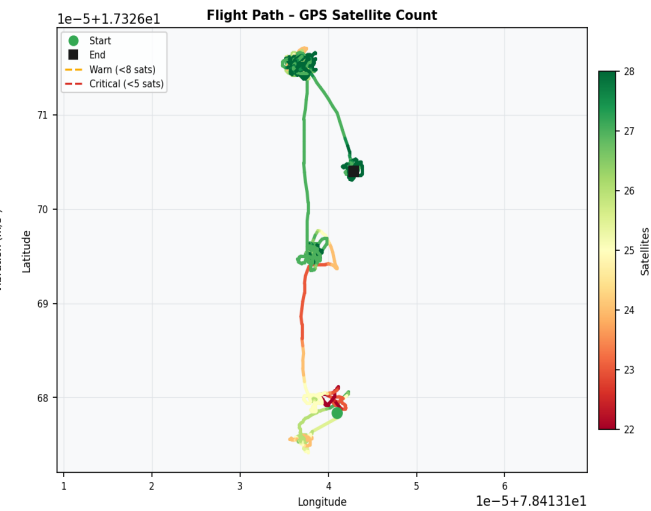
Flight Path – by Vibration

Path coloured by **peak vibration** (IMU 0). Green = calm → Red = severe.



Flight Path – by Satellite Count

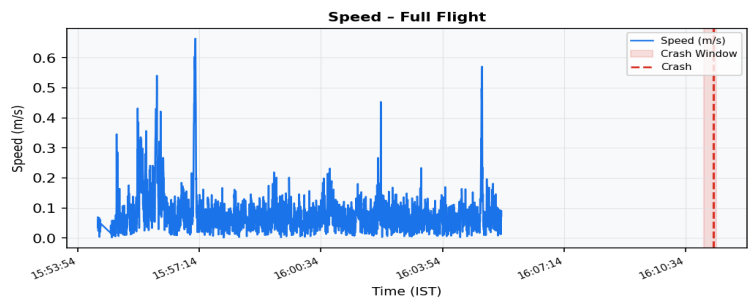
Path coloured by **GPS satellite count**. Green = strong fix → Red = weak fix.



Speed

PASS

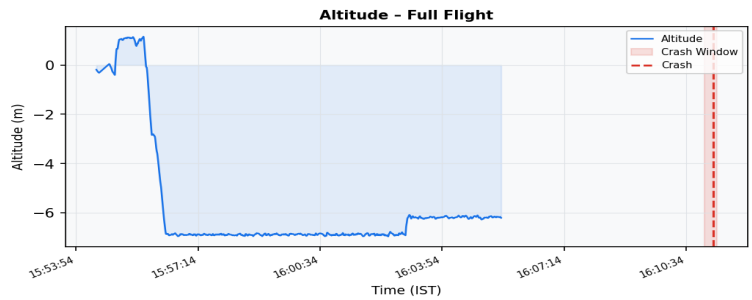
Speed normal: max=0.7 m/s



Height

PASS

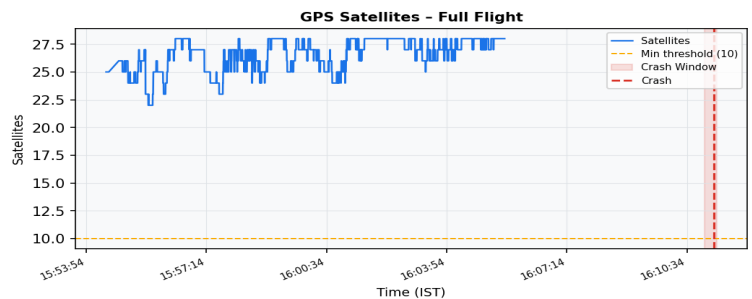
No abnormal descent
(max=1.2m, min=-7.0m)

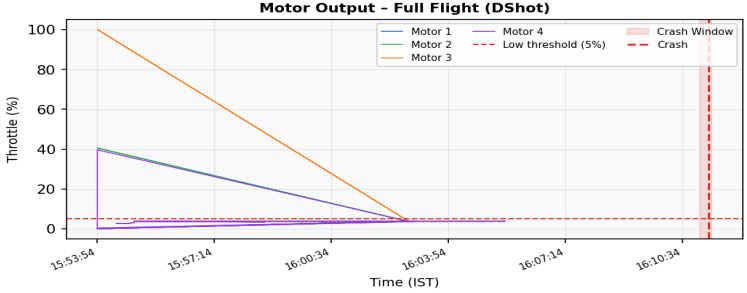
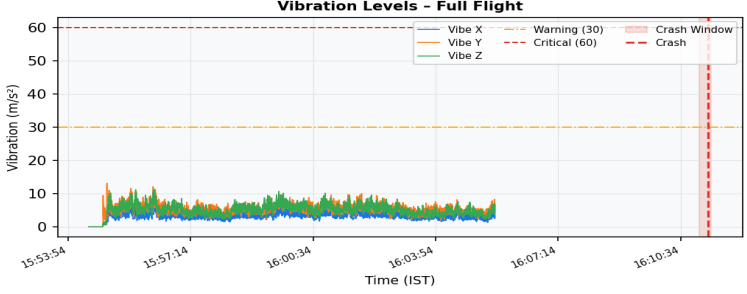
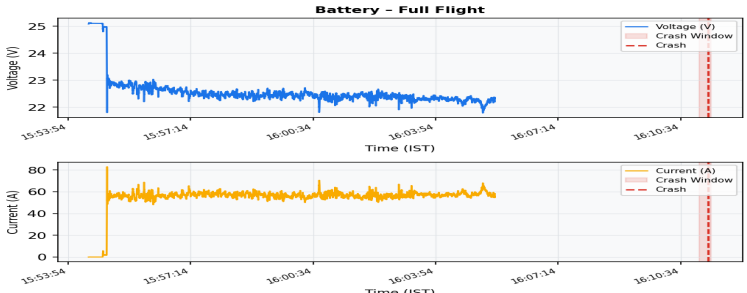
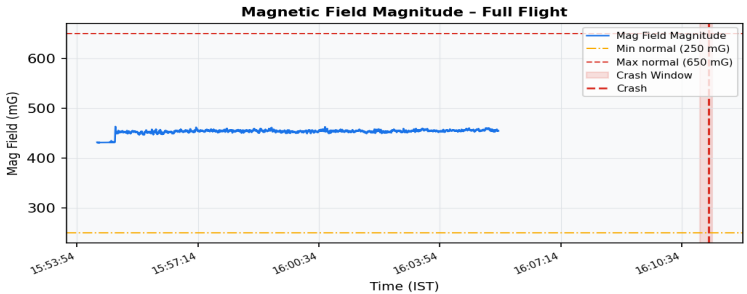
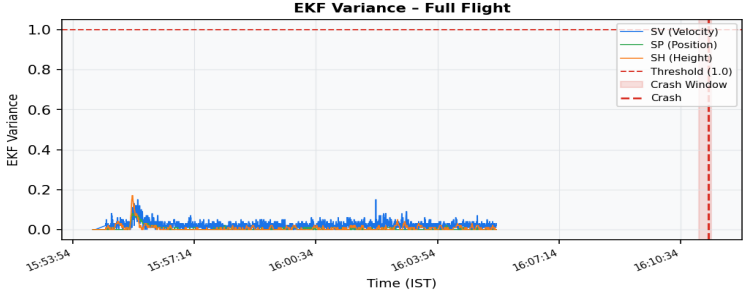


GPS Satellites

PASS

Satellite count good: min=22,
avg=27



Motor PWM	FAIL Motors at minimum before disarm (~670 DShot)	Motor Output - Full Flight (DSHOT) 
Vibration	PASS Vibration OK: X=8.9 Y=13.1 Z=11.2 m/s²	Vibration Levels - Full Flight 
Battery	PASS	Battery - Full Flight 
Magnetic Field	FAIL MAG0: avg=454mG min=431 max=463 cv=1% [PASS] MAG1: avg=799mG min=431 max=963 cv=5% [FAIL]	Magnetic Field Magnitude - Full Flight 
EKF Variance	PASS L0: SV=0.15 SP=0.08 SH=0.17 [PASS] L1: SV=0.15 SP=0.08 SH=0.18 [PASS] L2: SV=0.15 SP=0.08 SH=0.19 [PASS]	EKF Variance - Full Flight 

Flight Event Timeline

Real Time (IST)	Rel (+s)	Type	Event
15:54:28 IST	+33.3s	MODE	Mode → Loiter
15:54:48 IST	+54.1s	MODE	Mode → Loiter
15:54:50 IST	+55.8s	EV	Auto Armed
15:54:50 IST	+56.1s	EV	Motors Interlock Enabled
15:54:58 IST	+63.4s	EV	Not Landed (Taking Off)

Conclusion & Recommendations

Critical Issues Found:

- **IMU Health:** IMU100 gyro HW errors: EG max=1016957739 | IMU100 accel HW errors: EA max=991432351
- **Magnetic Field:** MAG0: avg=454mG min=431 max=463 cv=1% [PASS] | MAG1: avg=799mG min=431 max=963 cv=5% [FAIL]
- **Compass Interference:** EKF compass variance: SM max=163.49 (0% of flight > threshold 0.5)
- **RC Signal Loss:** RC signal gap(s) while armed: 1 event(s), longest 222.5s at t=63.8s (t=63.8s)
- **Crash Detection:** MSG: 'Logging @SYS/crash_dump.bin' (t=63.5s)
- **Motor PWM:** Motors at minimum before disarm (~670 DShot) (t=696.7s)
- **Crash Event:** MSG: 'Logging @SYS/crash_dump.bin' (t=63.5s)

Warnings:

- **Clean Boot:** No boot message found — log may be partial or format unrecognised
- **Power Stability:** Unloaded voltage drop: 0.21V at 3.2A (t=56.4s, in-flight) (t=56.4s)
- **IMU Vibration:** IMU2 high vibration: 50.0 m/s² (X:22.9 Y:18.5 Z:50.0)
- **EKF Yaw Alignment:** No yaw alignment completion message — GPS yaw or external heading may be in use, or log is partial

Root Cause Analysis:

The aircraft armed at 20-05-2026 15:54:50 IST and took off at 20-05-2026 15:54:58 IST. Flight duration (Armed → Disarmed): 981s (16.4 min). It landed/disarmed at 20-05-2026 16:11:19 IST (+1044.9s from log start, IST). ■■ **DATA CORRUPTION DETECTED:** Log terminated without disarm event (log stopped mid-flight) This indicates the log file is incomplete or was corrupted, most likely due to: (1) Crash impact causing SD card write failure, (2) Power loss/brownout during logging, (3) SD card ejection/failure mid-flight, or (4) Flight controller reset/crash. The absence of a proper disarm sequence and log termination strongly suggests an uncontrolled event (crash or power failure). Inspect the aircraft for physical damage and check SD card integrity. ArduPilot's internal crash checker triggered — the autopilot detected attitude or acceleration outside safe bounds and cut motor power. ■ IMU Health: IMU100 gyro HW errors: EG max=1016957739 | IMU100 accel HW errors: EA max=991432351 ■ Magnetic Field: MAG0: avg=454mG min=431 max=463 cv=1% [PASS] | MAG1: avg=799mG min=431 max=963 cv=5% [FAIL] ■ Compass Interference: EKF compass variance: SM max=163.49 (0% of flight > threshold 0.5) ■ RC Signal Loss: RC signal gap(s) while armed: 1 event(s), longest 222.5s at t=63.8s (t=63.8s) ■ Motor PWM: Motors at minimum before disarm (~670 DShot) (t=696.7s) ■ Crash Event: MSG: 'Logging @SYS/crash_dump.bin' (t=63.5s)